UNIT 4 · 9 hours

MOS Inverter and CMOS



Transistors

Basic Electrical and Electronics Engineering (EEE103)
Dr. Mitesh Astik · Dept. of Electrical and Electronics Engineering
NMAM Institute of Technology, Nitte · 2025-26

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Learning objectives

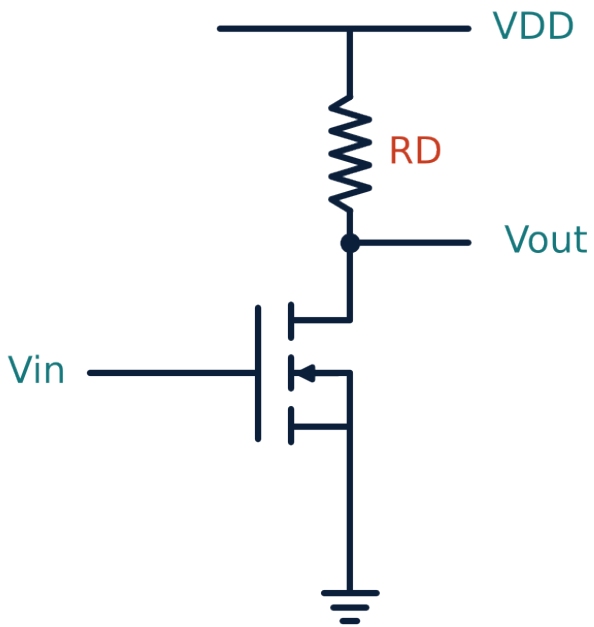
- Explain the operation of an NMOS inverter with a resistive load
- Explain the CMOS inverter with PMOS and NMOS
- Draw the voltage transfer characteristic
- State advantages of CMOS

Syllabus context

Construction and operation, BJT as a switch, working and characteristics of n-channel enhancement type MOSFET, common source amplifier, DC load line analysis, MOS inverter.

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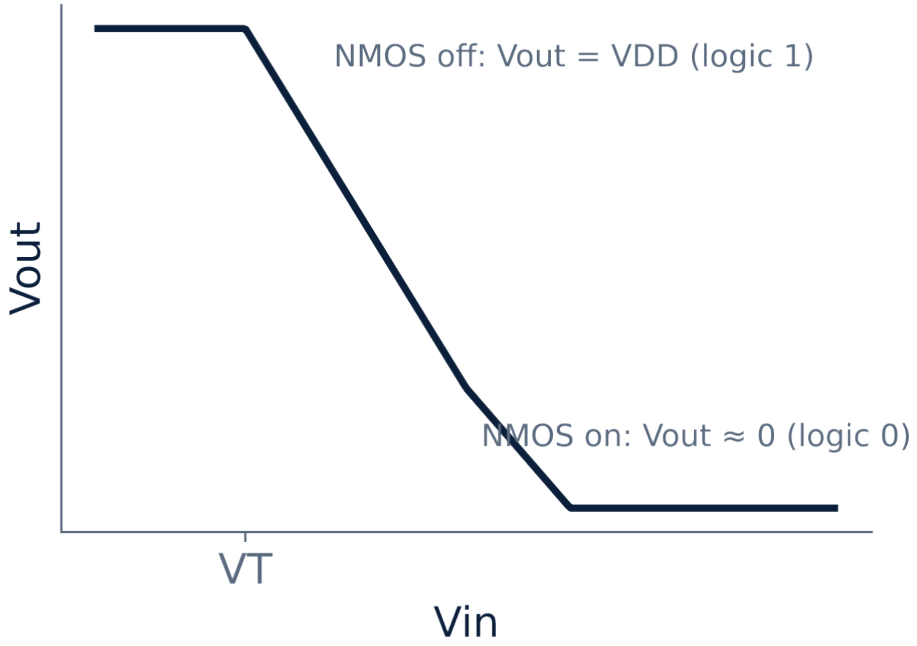
NMOS inverter (resistive load)



- Input at the gate, output at the drain; RD to VDD.
- Vin low (< VT): MOSFET off, Vout = VDD = logic 1.
- Vin high: MOSFET on, large current; Vout ≈ 0 = logic 0.
- Logic NOT function.

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Voltage transfer characteristic



- Flat at VDD until Vin reaches VT.
- Sharp transition region (high gain).
- Low output level for Vin above transition.
- Noise margins set by the transition voltages.

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CMOS inverter

Structure

A p-channel and an n-channel enhancement MOSFET on one substrate; gates joined as the input; drains joined as the output.

Vin = 5 V (1)

NMOS: VGS = 5 V, ON (low resistance). PMOS: VGS = 0 V, OFF. Output pulled to 0 V.

Vin = 0 V (0)

PMOS: VGS = −5 V, ON. NMOS: VGS = 0 V, OFF. Output pulled to 5 V.

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Truth table and device states (CMOS)

Vin	NMOS	PMOS	Vout
0 V (logic 0)	OFF	ON	5 V (logic 1)
5 V (logic 1)	ON	OFF	0 V (logic 0)

One transistor is always OFF, so there is no DC path from VDD to ground.

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NMOS resistive load versus CMOS

NMOS with RD

- One transistor + large resistor
- Static power dissipated when output is low
- Large area for resistor
- Asymmetric rise/fall

CMOS

- Complementary pair
- Almost zero static power
- Full logic swing 0 to VDD
- Basis of modern digital ICs

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Key takeaways

- MOS inverter implements logic NOT
- Resistive-load NMOS: Vin high → Vout low
- CMOS: PMOS + NMOS, only one ON at a time
- Very low static power; full-swing output

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Practice questions

Q1. Explain the working of a CMOS inverter.

Q2. Draw the VTC of an NMOS inverter.

Q3. Why does CMOS have almost zero static power?

Q4. State the output for Vin = 0 V in a 5 V CMOS inverter.